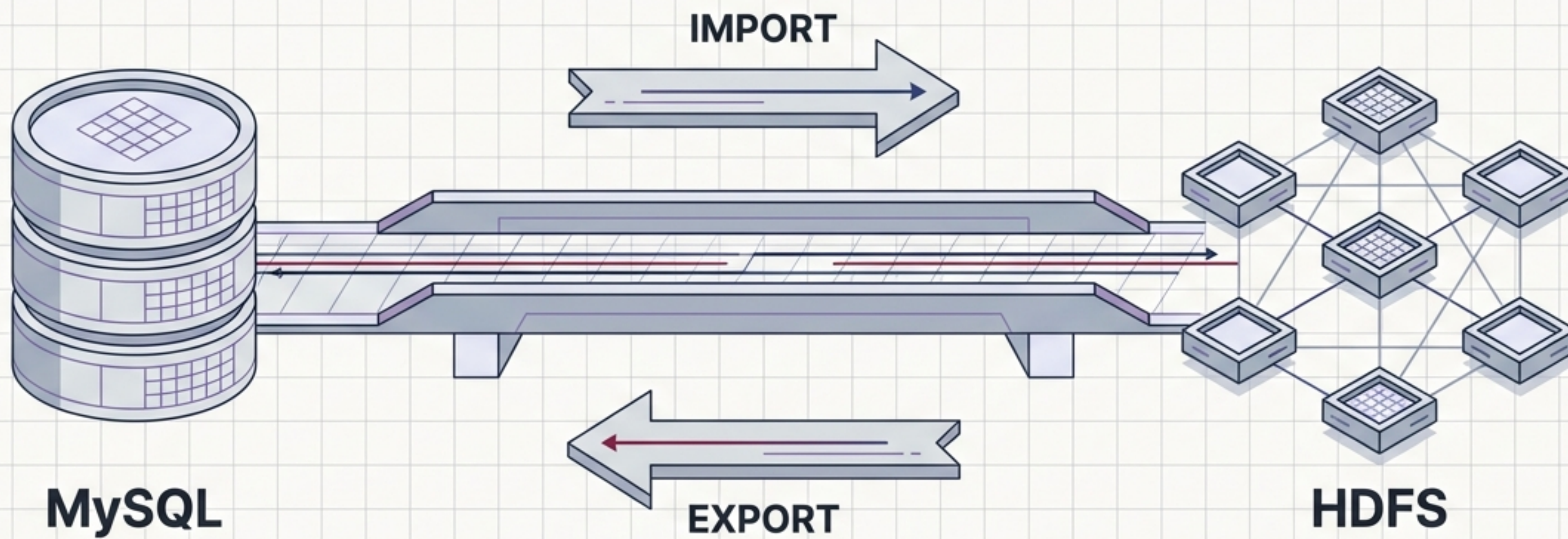


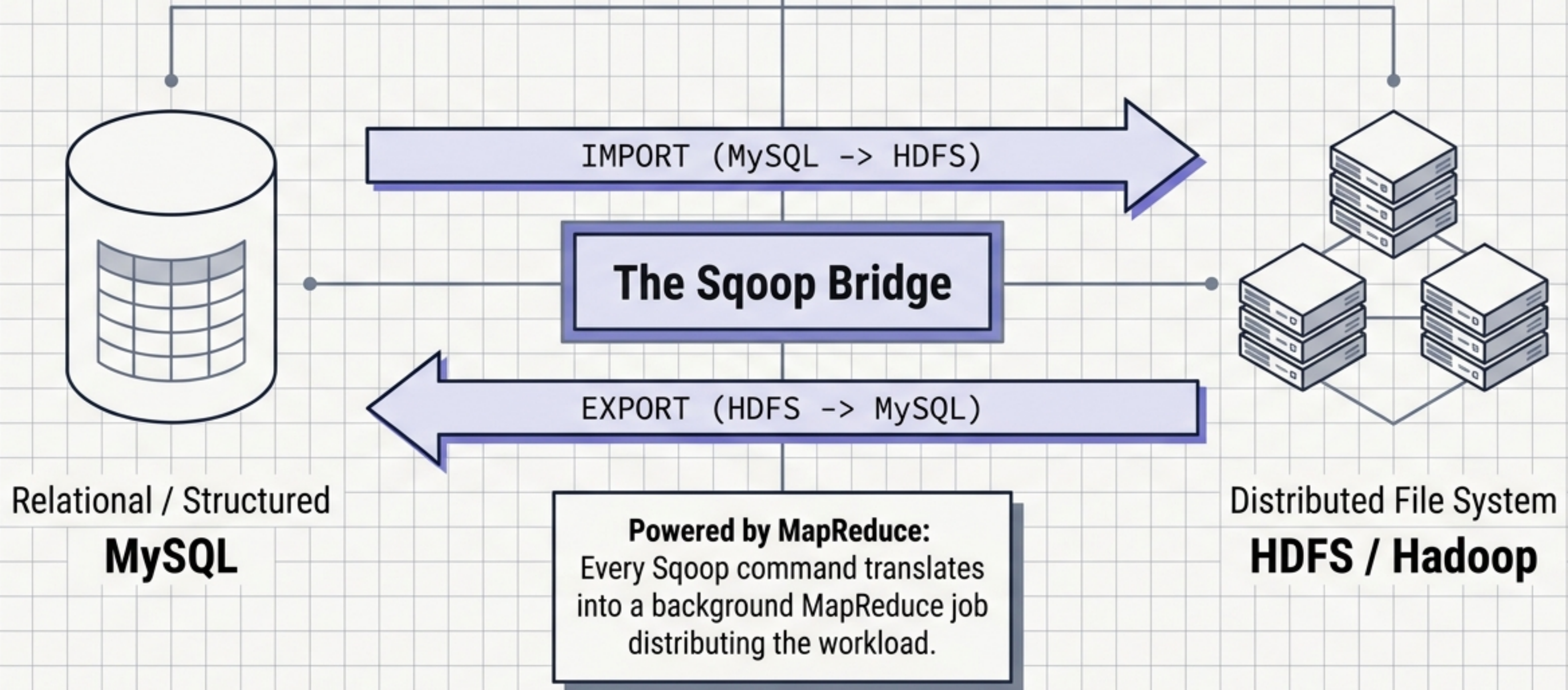
Apache Sqoop: The Data Engineer's Playbook

A definitive reference guide for installation, ingestion, egress, and automation.



ARCHITECTURE BLUEPRINT

PREREQUISITE: Hadoop MUST be running for any Sqoop command to execute.



Command Panel

```
sudo apt-get install mysql-server --fix-missing
```

```
mysql -u root -p
```

```
CREATE DATABASE exp;
```

```
USE exp;
```

```
CREATE TABLE employee (id INT, name VARCHAR(20));
```

```
INSERT INTO employee VALUES (1, 'Sample');
```

```
SELECT * FROM employee;
```

1. Installation

Use the --fix-missing flag if network issues cause library drops. Set root password when prompted.

2. Validation & Setup

Forcefully enter with root access and establish the test schema.

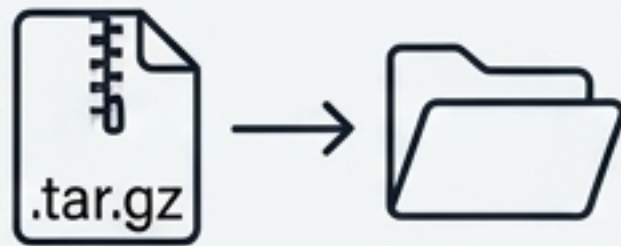
3. Verification

Verify insertion to ensure the database is ready for Sqoop integration. ✓

SQOOP CONFIGURATION WORKFLOW

01

Extract & Rename



Extract
sqoop-1.4.7.bin__hadoop.
tar.gz in the home directory.

Rename the verbose output
folder to sqoop to simplify
path configurations.

02

The MySQL Connector



Copy the MySQL JDBC
driver jar file directly into
the sqoop/lib directory.

This acts as the translation
layer between Sqoop and
MySQL.

03

Environment Template



Navigate to sqoop/conf.

Copy sqoop-env-
template.sh and rename
the new copy to
sqoop-env.sh.

SQOOP CONFIGURATION: ENVIRONMENT VARIABLES

`sqoop/conf/sqoop-env.sh`

Uncomment lines 23 & 26 and set absolute paths to your Hadoop installation.

```
export HADOOP_COMMON_HOME=/path/to/hadoop
export HADOOP_MAPRED_HOME=/path/to/hadoop
```

`~/.bashrc`

Run `sudo gedit $HOME/.bashrc`. Append Sqoop variables beneath existing Java/Hadoop paths, then run `source ~/.bashrc`.

```
export SQOOP_HOME=/path/to/sqoop
export PATH=$PATH:$SQOOP_HOME/bin
```

✓ **Validation:** Run '`sqoop version`' in the terminal to successfully print the installed 1.4.7 version.

Mapping the Source

```
sqoop list-databases --connect  
jdbc:mysql://localhost/ --username root -P
```

Returns default databases plus our newly created 'exp'.

```
sqoop list-tables --connect  
jdbc:mysql://localhost/exp --username root  
-P
```

Returns the 'employee' table inside the 'exp' database.

The --eval Feature

Concept: You don't need a separate MySQL terminal. Execute SQL directly through Sqoop to prep ingestion tests.

```
sqoop eval \  
  --connect jdbc:mysql://localhost/exp \  
  --username root -P \  
  --query "CREATE TABLE test_new (id INT);"
```

Quickly insert or create tables directly.

SQOOP COMMAND ANATOMY: IMPORT DIRECTIVE

```
sqoop import --connect jdbc:mysql://localhost/exp  
--username root -P --table test_new -m 1
```

The operation directive.

JDBC connection string targeting the 'exp' database.

Prompts for secure password (prevents plain text logging).

The specific MySQL source table to ingest.

Forces a single mapper task (ideal for small datasets).



Operational Note: Imports run as MapReduce jobs. If a job hangs for hours, copy the Job ID from the terminal and escalate to the Admin team for manual intervention.

The CLI Approach (Developer)

```
hdfs dfs -lsr /user/<username>/test_new
```

Checks directory creation.

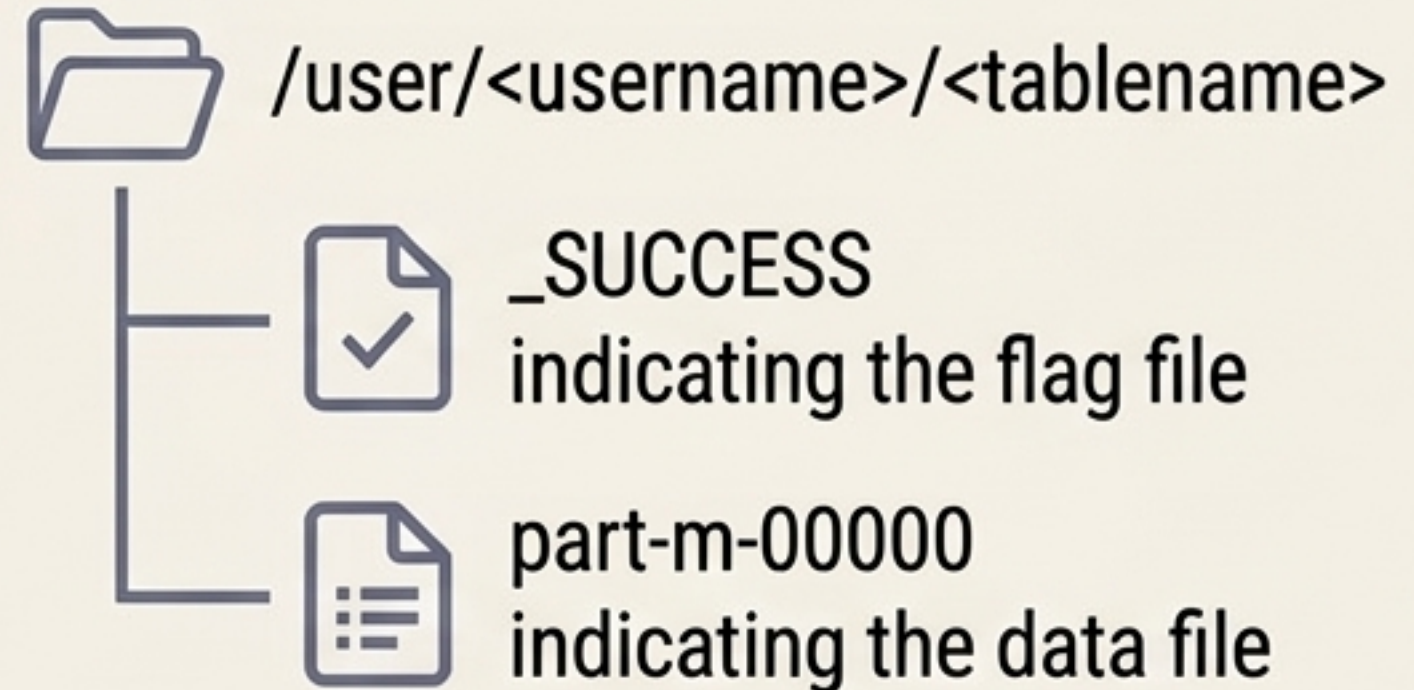
```
hdfs dfs -cat  
/user/<username>/test_new/part-m-00000
```

Reads the file to reveal actual data rows.

The UI Approach (Admin)

 localhost:50070

Utilities > Browse File System



SQOOP IMPORT STRATEGY MATRIX

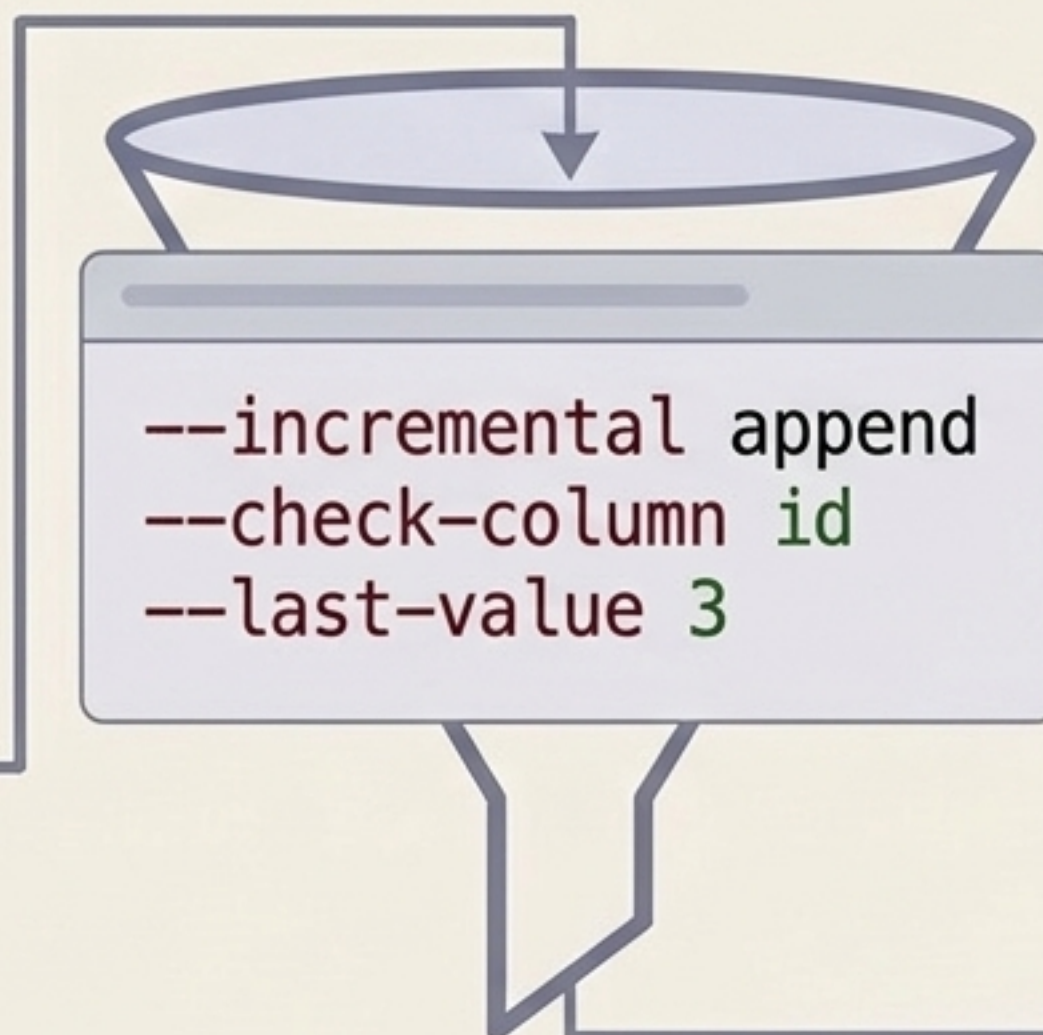
Full Table Import	Import-All-Tables	Conditional Import
<code>--table</code>	<code>sqoop import-all-tables</code>	<code>--where "id=3"</code>
Creates a target directory matching the table name. Fails if the directory already exists.	Iterates through every table in the DB. Generates separate MapReduce jobs per table. Skips tables with existing HDFS directories.	Filters records based on SQL logic. Crucial Rule: You MUST use the <code>--target-dir</code> flag to route filtered data to a fresh HDFS location to prevent collision errors with default directories.

INCREMENTAL APPEND PROCESS

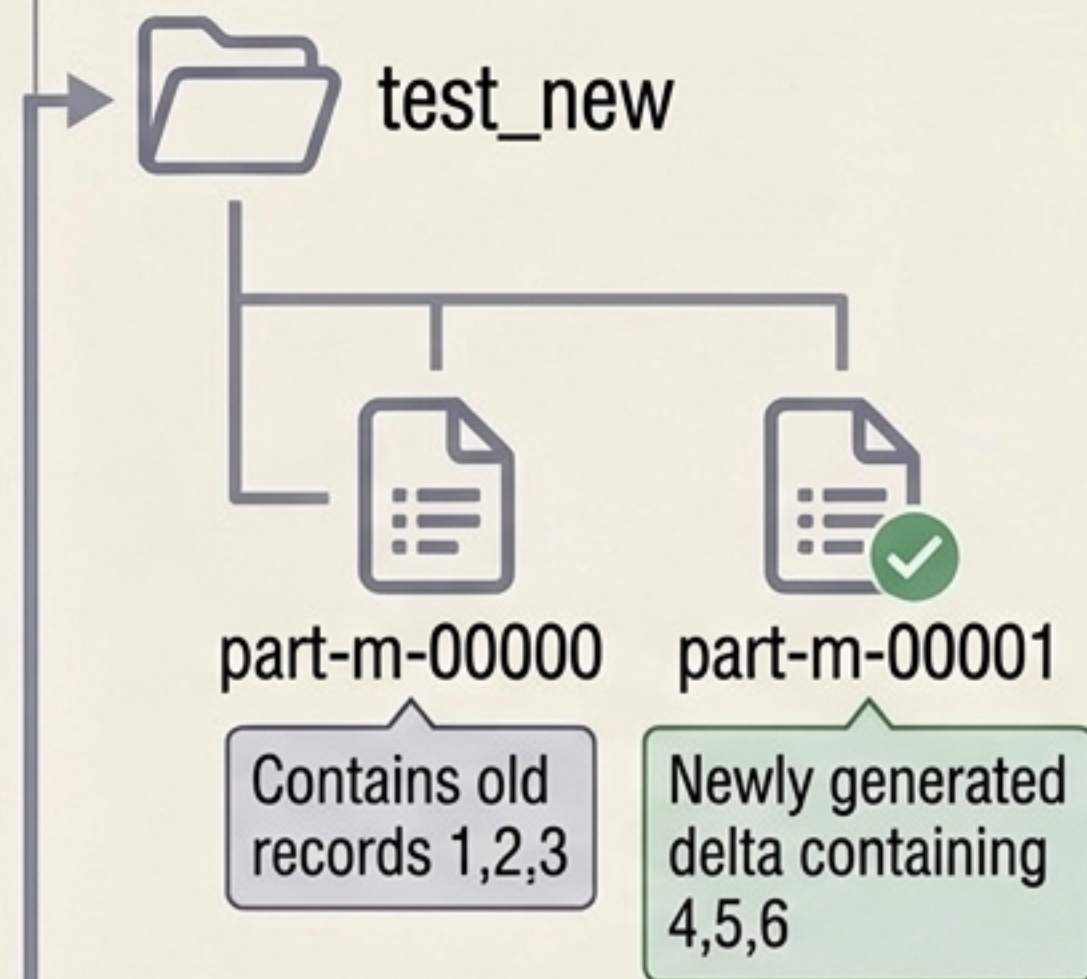
SOURCE: MySQL TABLE (with new inserts)

1	Record 1, Data A
2	Record 2, Data B
3	Record 3, Data C
4	Record 4, Data D
5	Record 5, Data E
6	Record 6, Data F

FILTER: INCREMENTAL APPEND LOGIC



TARGET: HDFS DIRECTORY (test_new)



Key Insight: Incremental appends do not overwrite existing HDFS files; they generate new part files within the same directory containing only the delta records.

DATA EGRESS: EXPORTING HDFS TO MYSQL



WARNING: Unlike Import, Sqoop Export DOES NOT create the target table. **Columns must exactly match** the HDFS file schema, or export fails.

1. The Source (HDFS)

```
hdfs dfs -put  
license.txt  
/exp_data
```

Push local text file
to HDFS path.

2. The Prerequisite

```
CREATE TABLE  
personal_info  
id 207,  
column CREE,  
value NULL  
);
```

Create an empty
table in MySQL.

3. The Execution

```
sqoop export  
--connect...  
--table  
personal_info  
--export-dir  
/exp_data
```

Trigger the egress
mapping.

4. Validation

Check MySQL to
confirm records
exist in
personal_info.

Check MySQL to
success.



Creating the Job

```
sqoop job --create myjob  
-- list-databases  
--connect...
```

The ‘--’ acts as a delimiter. Everything after it is the functional query saved under the alias ‘myjob’.

Executing the Job

```
sqoop job --exec myjob
```

Triggers the saved command. Prompts for the MySQL password dynamically to ensure security.

Automation Note: You can create multiple saved jobs (e.g., myjob2, myjob3) for different operational tables and execute them via cron or scheduling tools.


```
sqoop job --exec myjob
```

JDBC connection exception: missing connector.



Sqoop Jobs require an additional JSON library for metadata execution that is not included in the default tarball.

1. Locate `java-json.jar`
2. Copy it directly into the `sqoop/lib` directory alongside the MySQL connector.
3. Run `source ~/.bashrc` to refresh.
4. Re-execute the job successfully.

Sqoop Command Synthesis



Infrastructure & Admin	Ingestion (Import)
<pre>sqoop version sqoop list-databases sqoop list-tables sqoop eval --query "..."</pre>	<pre>sqoop import --table <name> sqoop import-all-tables sqoop import --table <name> --where "<condition>" \ --target-dir <path> sqoop import --incremental append \ --check-column <col> --last-value <val></pre>
Egress (Export)	Automation (Jobs)
<pre>sqoop export --table <name> \ --export-dir <hdfs_path></pre>	<pre>sqoop job --create <name> -- <command> sqoop job --exec <name></pre>